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Development of the ReaxFF Reactive Force Field for Inherent Point Defects in the Si/Silica System

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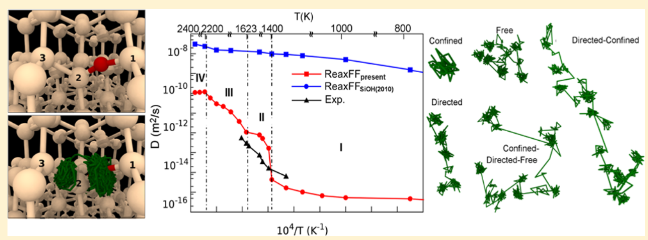
Supporting Information

ABSTRACT: We redeveloped the ReaxFF force field parameters for Si/O/H interactions that enable molecular dynamics (MD) simulations of Si/SiO_2 interfaces and O diffusion in bulk Si at high temperatures, in particular with respect to point defect stability and migration. Our calculations show that the new force field framework ($ReaxFF_{present}$), which was guided by the extensive quantum mechanical-based training set, describes correctly the underlying mechanism of the O-migration in Si network, namely, the diffusion of O in bulk Si occurs by jumping between the neighboring bond-centered sites along a path in the (110) plane, and during the jumping, O goes through the asymmetric transition state at a saddle point. Additionally, the $ReaxFF_{present}$ predicts the diffusion barrier of O-interstitial in the bulk Si of 64.8 kcal/mol, showing a good agreement with the experimental and density functional theory values in the literature. The new force field description was further applied to MD simulations addressing O diffusion in bulk Si at different target temperatures ranging between 800 and 2400 K. According to our results, O diffusion initiates at the temperatures over 1400 K, and the atom diffuses only between the bond-centered sites even at high temperatures. In addition, the diffusion coefficient of O in Si matrix as a function of temperature is in overall good agreement with experimental results. As a further step of the force field validation, we also prepared amorphous SiO_2 (a-SiO_2) with a mass density of 2.21 gr/cm^3, which excellently agrees with the experimental value of 2.20 gr/cm^3, to model a-SiO_2/Si system. After annealing the a-SiO_2/Si system at high temperatures until below the computed melting point of bulk Si, the results show that $ReaxFF_{present}$ successfully reproduces the experimentally and theoretically defined diffusion mechanism in the system and succeeded in overcoming the diffusion problem observed with $ReaxFF_{SiOH(2010)}$, which results in O diffusion in the Si substrate even at the low temperature such as 300 K.



1. INTRODUCTION

Many crystal growth methods in the semiconductor industry rely on Si and its interface with silica because of remarkable synergy between Si and silica (a-SiO_2).[1−5] Therefore, several studies have attempted to gain deeper insight into the formation of the a-SiO_2/Si interface which is accomplished by oxygen diffusion process during Si oxidation.[6−8] Under the thermal treatment of the interface, an O atom moves through the silica network by leaving a vacancy behind and reacts with Si−Si bond in the vicinity of the interface. The O atom residing in Si−Si bond-centered (BC) site continues to diffuse along a path comprising a combination of O jumps between BC sites in the Si substrate.[9−12] During the jump, the O atom goes through transition state by overcoming the diffusion barrier of 2.53 eV.[13,14] Recent density functional theory (DFT) studies[10,11] showed the dependence of the energy barrier on the geometrical description of the transition state. According to the DFT reports,[10,11] asymmetric transition pathway yields almost the same energy barrier with the experimental values,[13,14] whereas symmetric diffusion concept considered by most of the earlier studies[15−17] leads to a significantly underestimated energy barrier value.

Here, we present a new computational framework to atomistically model the temperature dependence of O diffusion in bulk Si and Si/SiO_2 interface, for which we reparameterized the ReaxFF force field parameters developed by Fogarty et al. ($ReaxFF_{SiOH(2010)}$)[18] for Si/O/H interactions. In line with DFT works,[13,14] we expanded the training set taken from ref 18 by including the formation energies of O-related point defects in a-SiO_2/Si system and minimum energy migration pathway of O in bulk Si. The new force field description was then validated with the two-phase molecular dynamics (MD) simulations for the determination of the Si melting point, MD simulations of the temperature dependence of O diffusion in bulk Si, and finally, annealing of a-SiO_2/Si system at temperatures ranging between room temperature

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and the melting point of Si. The rest of the paper is divided into two main sections: in Section 2, we present the computational details of the force development and MD simulations together with the results and discussion, and Section 3 is devoted to our concluding remarks.

2. METHODOLOGY, RESULTS AND DISCUSSION

2.1. ReaxFF Reactive Force Field. The ReaxFF reactive force field developed by van Duin et al.[19,20] contributes to bridging between quantum mechanical (QM) and the classical MD methods. ReaxFF enables successful analysis of the dynamic behaviors of larger and chemically reactive systems on the atomic level, whereas QM methods are limited to systems including a few atoms because of the complexity of analytically solving Schrodinger equation for the many-body wave function of the electrons. Contrary to the static bond assumption in classical MD simulations, which enforces fixed bonding, ReaxFF is based on the relationship between bond order and bond distance updated at every MD step.[21−24] It can modify the bond strength depending on the environment by breaking and reforming bonds during simulations. This feature allows ReaxFF to produce a realistic, geometry-dependent charge distribution. The total energy of the system is expressed as a sum of partial contributions of bond order-dependent terms, such as bond, valence angle, torsion, and nonbonded van der Waals and Coulomb interactions which are calculated between every pair of atoms, regardless of connectivity. For the van der Waals and Coulomb energy descriptions, ReaxFF uses a shielded Morse-potential and the electronegativity equalization method[25] to define the short-range interactions and polarization/charge transfer effects, respectively. More details about the ReaxFF force field formalism are given in earlier publications.[19,20,22]

2.2. Force Field Parametrization. The Si/O force field ($\mathrm{ReaxFF_{SiO(2003)}}$) has first developed by van Duin et al.[20] and applied successfully to silicon and silica systems and Si/SiO_2 interfaces at the temperatures up to 900 K.[26−31] Then, the ReaxFF force field framework was further extended to Si/O/H interactions ($\mathrm{ReaxFF_{SiOH(2010)}}$) by Fogarty et al.,[18] which successfully describes proton-transfer reactions at the water/silica interface. However, $\mathrm{ReaxFF_{SiOH(2010)}}$ is unable to simulate Si/SiO_2 interface at high temperatures. This inability can be attributed to inadequate data description incorporated in the training set which initially contains the QM-based equations of state for condensed phases of Si and SiO_2 and structures and energy barriers for SiO_2 clusters. We, therefore, included the formation energies of O-related point defects in a-SiO_2/Si system, and O-migration pathway in bulk Si to accurately treat the Si/SiO_2 interface at high temperatures, essentially, any temperature below the computed melting temperature of bulk Si. Using this augmented training set, we refitted the force field parameters by using a single-parameter search optimization method. With this method, the parameters were determined by minimizing the sum of the squared error between ReaxFF values and QM/experimental data until ReaxFF values and the QM/experiment data reached a compromise during the training, as shown eq 1

$$\text{error} = \sum_{i=1}^{n}\left[\frac{(x_{i,\mathrm{QM/Exp.}} - x_{i,\mathrm{ReaxFF}})}{\sigma}\right]^2 \tag{1}$$

where $x_{\mathrm{QM/Exp.}}$ and x_{ReaxFF} are the QM/experimental and the ReaxFF calculated value, respectively. σ is the weight factor determined in the training set. During the optimization, only the Si−Si bond, Si−O bond and off-diagonal, and Si-related valence angle parameters of the force field were modified. The following subsections present the details of QM data included in the training set from the earlier DFT publications[9−11,32] and our force field parametrization results.

2.2.1. Point Defects. Point defects play a profound role in the kinetics of diffusion and oxidation reactions observed in solid phases. Diffusion of the light elements (or their vacancies) such as O occurs through local point defects in the structure, in our case, through the bond-centered interstitial sites in both Si[9−11] and a-SiO_2[33] matrices at the a-SiO_2/Si interface. The following reaction describes the mechanism underlying the diffusion of O observed at the interface.

$$\mathrm{Si} + \mathrm{SiO_2} \rightarrow \mathrm{SiO}_x + \mathrm{SiO}_{2-x} + \mathrm{V_O} \tag{2}$$

According to this reaction, an O atom coming from a-SiO_2 leaves a vacancy behind in a-SiO_2 which leads to an interstitial defect formation in Si network by binding to a host Si atom. In the training set, we first described the key elements of the O diffusion through the a-SiO_2/Si interface, which are an O-vacancy in a-SiO_2[32] and O-interstitial types in bulk Si.[9] Subsequently, we included the minimum energy migration pathway of the O atom in bulk Si.[10,11]

2.2.2. O-Vacancy in a-SiO_2. O-vacancies were created in bulk a-SiO_2, containing 216 atoms, by randomly removing 18 O atoms from the structure. The Cartesian coordinates of the a-SiO_2 supercell were taken from ref 34. The following formula was defined in the training set for the description of the vacancy formation energy

$$E_{\mathrm{form,vac}} = 1/n \times [E_{(\mathrm{silica}-n\mathrm{O})} - (E_{\mathrm{silica}} - n/2E_{\mathrm{O_2}})] \tag{3}$$

where n is the number of vacancies in the supercell, 18, $E_{\mathrm{silica-nO}}$ is the total system energy of the supercell including 18 vacancies, and E_{silica} and $E_{\mathrm{O_2}}$ are the reference energies of bulk amorphous silica containing 216 atoms and O_2 molecule in a vacuum. Table 1 presents the formation energy of an O-vacancy predicted by $\mathrm{ReaxFF_{present}}$, DFT,[32] $\mathrm{ReaxFF_{SiO(2003)}}$, and $\mathrm{ReaxFF_{SiOH(2010)}}$. $\mathrm{ReaxFF_{present}}$ correctly predicts the vacancy formation energy with only a slight difference from DFT.[32]

Table 1. Formation Energy (in kcal/mol) of O-Vacancy in a-SiO_2 Computed by DFT[32] and $\mathrm{ReaxFF_{present}}$, $\mathrm{ReaxFF_{SiOH(2010)}}$, and $\mathrm{ReaxFF_{SiO(2003)}}$[a]

defect	DFT	$\mathrm{ReaxFF_{present}}$	$\mathrm{ReaxFF_{SiOH(2010)}}$	$\mathrm{ReaxFF_{SiO(2003)}}$
$E_{\mathrm{form,vac}}$	139.0	134.5	132.2	115.4

[a] $E_{\mathrm{form,vac}}$ is the formation energy of O-vacancy in a-SiO_2.

2.2.3. O-Interstitial in Diamond Si. Interstitial atoms residing in highly symmetric sites in diamond Si have been commonly reported as bond centered (BC) between two neighboring host Si atoms, hexagonal (H), tetrahedral (T), and split (S) in the literature.[9,35−38] In the Hermann−Mauguin notation, the point group of T site is 43m, that of BC and H is 3m, and point group of S site switches between 42m and 3m depending on the atom type.[37] Figure 1 illustrates the first three interstitial configurations in diamond Si unit cell. The last one, S site, is depicted in the next section. The fractional coordinates of O atoms at BC, T, and H sites within a

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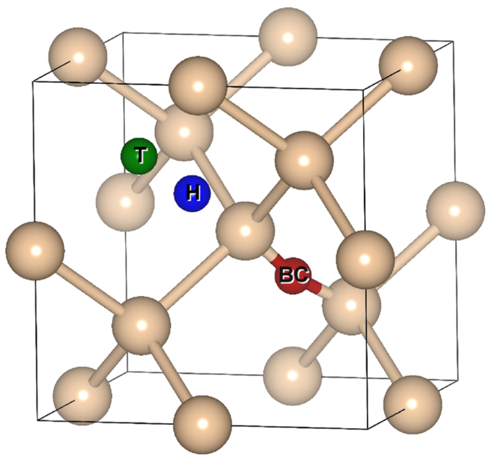

Figure 1. Ball–stick representation of O-interstitial configurations in the unit cell of Si where green, blue, and red represent T, H, and BC sites, respectively (Figures are generated by VESTA.[39] From now on, the figures generated by VESTA will be notified only by VESTA).

conventional unit cell are (0.612, 0.388, 0.354), (0.25, 0.75, 0.75), and (0.375, 0.625, 0.625), respectively.

According to refs9–11 the BC site is the most stable followed by the S, T, and H sites, as indicated in Table 2 where

Table 2. Relative Total Energies for Four Different O-Interstitial Configurations in Bulk Si Computed by ReaxFF$_{present}$, ReaxFF$_{SiOH(2010)}$, ReaxFF$_{SiO(2003)}$, DFT,[9] and Experiments[13,14]

interstitial type	BC (kcal/mol)	H (kcal/mol)	T (kcal/mol)	S (kcal/mol)
Exp.	0			58.8
DFT	0	87.9	85.8	57.3
ReaxFF$_{present}$	0	85.4	84.4	64.8
ReaxFF$_{SiOH(2010)}$	0	−5.4	15.1	52.6
ReaxFF$_{SiO(2003)}$	0	48.7	84.2	36.9

the formation energy of the O atom at the BC position is taken as a reference. However, ReaxFF$_{SiOH(2010)}$ is unable to reproduce the relative stability of the O-interstitial defects leading to an undesirable O diffusion even at low temperatures as seen in Figure 2 where O atoms, though being initially positioned at the BC sites in the matrix containing 512 Si and

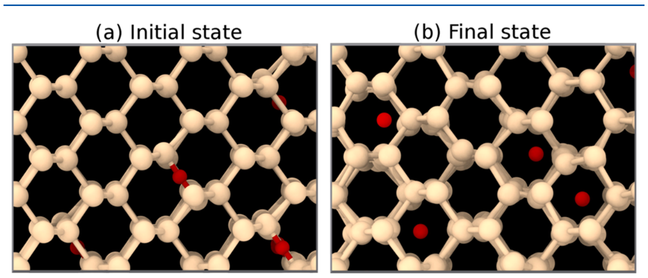

Figure 2. Snapshots taken from the MD simulation performed by ReaxFF$_{SiOH(2010)}$ at 1000 K. (a) The initial configuration of the supercell where O atoms are initially positioned at BC sites and (b) the annealed supercell at 1000 K for 2.5 ns where O atoms are positioned at H and T sites inside the units of honeycomb in the lattice (This visualization was performed using the OVITO software.[40] From now on, the figures generated by OVITO will be notified only by OVITO).

15 O atoms (Figure 2a), diffuse through a jump sequence of H → T → H sites (Figure 2b) instead of the BC → S → BC sequence under the heat treatment at 1000 K.

In the present study, to accurately predict the relative stability order of the interstitial sites in the bulk Si and thus, to remedy the O diffusion problem observed in the Si matrix by ReaxFF$_{SiOH(2010)}$, four supercells were constructed from the diamond Si unit cell with a lattice constant of 5.463 Å. Each supercell with a dimension of 16.4 Å contains 216 Si atoms and an additional O atom individually positioned at the BC, H, T, and S sites. The following energy formula was introduced in the training set for the calculation of the interstitial formation energy

$$E_{form,ins} = E_{(silicon+O)} - \left(E_{silicon} + \frac{1}{2}E_{O_2}\right) \tag{4}$$

where $E_{(silicon+O)}$ is the total energy of the system consisting of bulk Si with an additional O atom, $E_{silicon}$ and E_{O_2} are the reference energies of the bulk Si and O_2 molecule, respectively. After refitting the force field parameters of interest, ReaxFF$_{present}$ reproduced the relative stability of interstitial configurations in the same order as DFT,[9] as evident from Table 2. The BC site was successfully determined to be the most stable, and the O atom is allowed to diffuse by a correct jump sequence of BC → S → BC.[9−11,35,36] Furthermore, the relative formation energies of H and T sites with respect to the BC site are estimated to be very close to the DFT values with differences of 2.5 and 1.4 kcal/mol, respectively. Additionally, ReaxFF$_{present}$ obtained a good agreement with DFT/experiment by predicting the relative energy of the S site, which is effectively the energy barrier, to be 64.8 kcal/mol.

2.2.4. Migration of O-Interstitial in Diamond Si. Recent DFT studies[10,11] demonstrated that an asymmetric transition pathway of O atom in diamond Si (Figure 3a) yields a more accurate result than the earlier DFT studies,[15−17] which considered a symmetric transition state (Figure 3b).

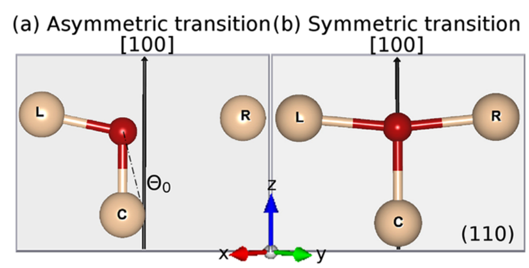

Figure 3. Symmetric and asymmetric transition states of O-interstitial in Si where L, R, C, and Θ_0 located on (110) plane represent the Si atoms at left and right sides with respect to [100] axis, the center Si atom and angle of O atom with respect to the [100] direction, respectively (VESTA[39]).

In this study, the migration path of O atom involving BC and S sites and the (110) plane was derived by taking into account the asymmetric transition concept (Figure 3a). Eight supercells were built from diamond Si unit cell. An additional O atom was individually defined in each supercell containing 216 Si atoms, as illustrated in Figure 4. The calculations were carried out by applying bond restraint only to the Si–O bond distances and keeping relaxed the remaining internal coordinates for the intermediate points. For only the BC (Figure 4a,h) and S (Figure 4e) configurations, bond restraint

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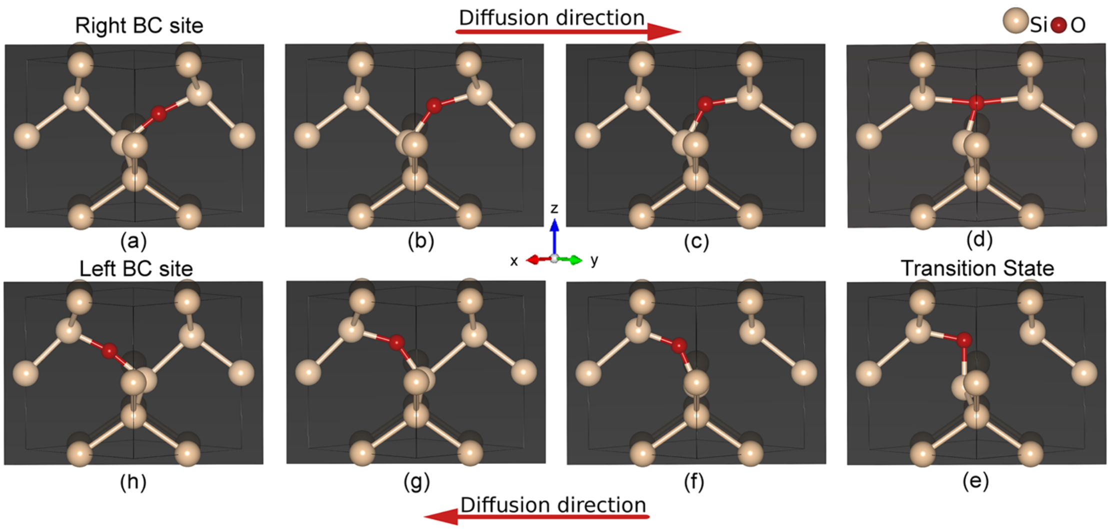

Figure 4. Ball−stick representation of minimum energy migration path of O-interstitial in the Si unit cell, derived from refs10, 11 as an input data for the force field training set where the gray background represents the (110) plane in the [100] direction. O diffuses by jumping from (a) the right BC site to (h) the left BC site or vice versa. During the diffusion, (e) O atom goes through the asymmetric transition state at the S site (VESTA[39]).

was also applied to the Si−Si bond distances between the center and right and the center and left Si atoms. Note that all three Si and O atoms were positioned on the (110) plane directed along the [100] axis as represented in Figure 3a, and the presence of O atom neighboring two host Si atoms causes slight distortions of the Si atoms from their ideal sites because of the formations of Si−O bond and Si−O−Si angle between them.[10,11]

The diffusion barrier that needs to be overcome by O atom to achieve the transition state at the S site was calculated as 64.8 by $\text{ReaxFF}_{\text{present}}$ which is in qualitative agreement with the experimental[13,14] and DFT[9] values with a slight shift by 6 and 7.5 kcal/mol, respectively, as shown in Table 2 where the barrier is a simply energy difference between the S and BC sites. Also, the graph displayed in Figure 5 shows that

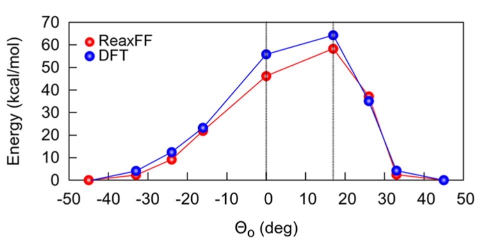

Figure 5. Energy of O-interstitial diffusing between two neighboring BC sites in the (110) plane in the Si matrix, estimated by QM[10] and $\text{ReaxFF}_{\text{present}}$. Θ_0 is the angle of O atom with respect to the [100] axis shown in Figure 3. $\text{ReaxFF}_{\text{present}}$ produces an asymmetric transition state at $\Theta_0 = 17°$ with an energy barrier of 64.8 kcal/mol, showing good agreement with DFT.[11]

$\text{ReaxFF}_{\text{present}}$ successfully resulted in an asymmetric migration path for O atom, in line with DFT reports.[10,11] Note that $\text{ReaxFF}_{\text{SiO(2003)}}$ predicts correctly BC site to be the most stable site, as well. However, it produced a significantly lower energy barrier of 36.9 kcal/mol as displayed in Table 2 and resulted in a symmetric transition state. The lower energy barrier can be attributed to the geometry of the migration pathway, as reported in DFT works.[10,11]

Figures S1 and S2 show the comparison between $\text{ReaxFF}_{\text{present}}$ and QM energy/volume curves for the equations of state for different polymorphs of Si and SiO_2 condensed phases. Additionally, the reaction energies of Si/O/H molecules computed by $\text{ReaxFF}_{\text{present}}$ and DFT[41] are displayed in Table S1. $\text{ReaxFF}_{\text{present}}$ force field parameters developed in this work are supplied in the Supporting Information material.

2.3. ReaxFF MD Simulations. *2.3.1. Predicting Melting Temperature of Bulk Si by Two-Phase Approach.* A two-phase simulation method was employed to accurately to determine the melting point of bulk Si based on $\text{ReaxFF}_{\text{present}}$ and $\text{ReaxFF}_{\text{SiOH(2010)}}$. A supercell including 5488 Si atoms was constructed from a diamond unit cell, established in a periodic $38.3 \times 38.3 \times 76.5$ Å^3 computational box. The half part of the supercell in the *z*-direction was frozen, and the other part was annealed in an NVT ensemble by setting their initial temperatures at 300 and 5000 K, respectively, to comprise two phases in co-existence (Figure 6a). Each of the solid and liquid parts has the same density and contains 2744 atoms. As a next step, the system was further equilibrated at the target temperature in an NPT ensemble to determine which of two phases grows during the simulation. When the temperature was above the melting point of Si, the solid−liquid interface moved toward to the crystal phase (Figure 6b) or vice versa (Figure 6c), and the most stable phase eventually filled the entire volume of the computational box (Figure 6d,e). This process was repeated at different temperatures by applying both versions of the force field, $\text{ReaxFF}_{\text{SiOH(2010)}}$, and $\text{ReaxFF}_{\text{present}}$, until reaching the melting temperature of bulk Si.

Figure 6 shows the solid and liquid phases of Si generated by $\text{ReaxFF}_{\text{present}}$. Our current force field parametrization resulted in the crystal phase completely overtaking the liquid phase at 2421 K (Figure 6d), whereas the liquid phase filled the volume of the box at 2425 K (Figure 6e). The melting point of bulk Si was determined to be a temperature between 2421 and 2425 K. $\text{ReaxFF}_{\text{SiOH(2010)}}$ yielded the melting temperature between 2689 and 2690 K.

Table 3 presents the data belonging to the structural properties of liquid phases of Si at 2425 and 2690 K, by $\text{ReaxFF}_{\text{present}}$ and $\text{ReaxFF}_{\text{SiOH(2010)}}$, respectively, and solid phase

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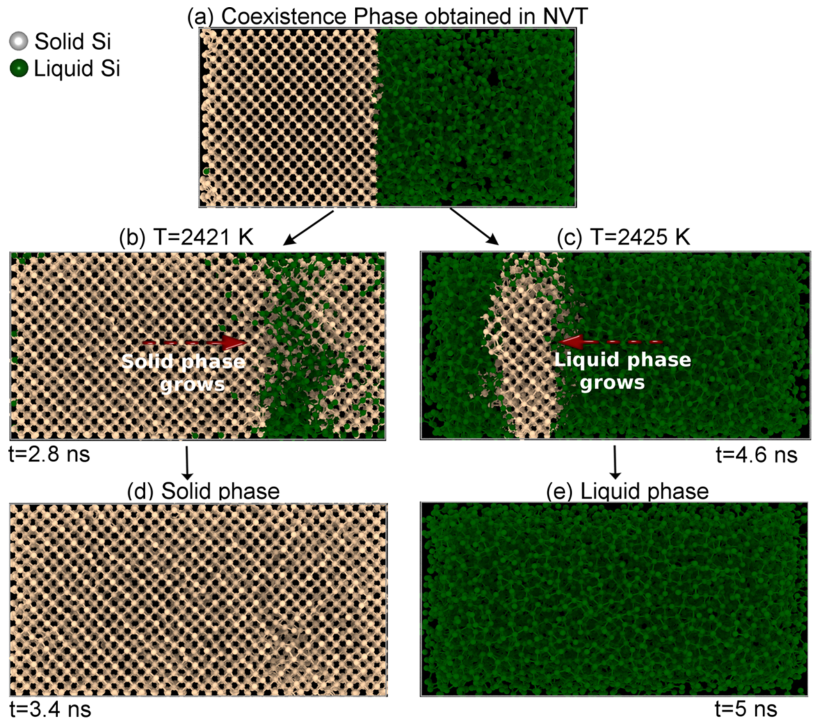

Figure 6. Snapshots taken from the two-phase MD simulations. (a) Co-existence phase including liquid and solid of Si obtained in the NVT, (b) solid/liquid interface moving toward the liquid phase, (c) solid/liquid interface moving toward the crystal phase, (d) crystal phase completely overtaking liquid phase at 2420 K for 3.4 ns, (e) liquid phase filling the box at 2421 K for 5 ns (OVITO[40]).

Table 3. Structural Properties of the Solid and the Liquid Phases of Si Generated by ReaxFF$_{SiOH(2010)}$ and ReaxFF$_{present}$ and their Experimental Values

property		ReaxFF$_{SiOH(2010)}$	ReaxFF$_{present}$	experiment
melting point (K)		(2689−2690)	(2421−2425)	[a]1683
density (gr/cm^3)	α-Si	2.48	2.47	[b]2.33
	liquid Si	2.52	2.61	[c]2.56, [d]2.58
				[e]2.61, [f]2.65
d_{Si-Si} (Å)	α-Si	2.296	2.312	[g]2.368
	liquid Si	2.392	2.376	

[a]Ref 42. [b]Ref 43. [c]Ref 44. [d]Ref 45. [e]Ref 46. [f]Ref 47. [g]Ref 48.

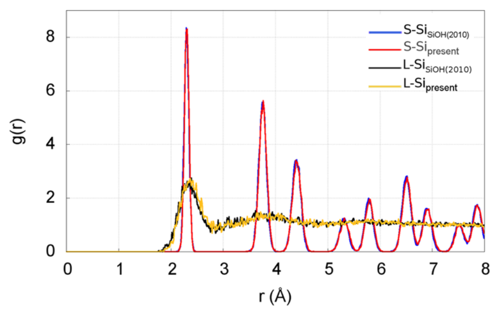

Figure 7. Partial RDFs belonging to the liquid and solid phases of Si. L-Si$_{SiOH(2010)}$ and S-Si$_{SiOH(2010)}$ represent liquid and solid phases generated by ReaxFF$_{SiOH(2010)}$. In the same manner, L-Si$_{present}$ and S-Si$_{present}$ represent the solid and liquid phases produced by ReaxFF$_{present}$.

of Si at 300 K by, again, both versions of ReaxFF. The density of the solid phase is still overestimated by ReaxFF$_{present}$, but the density of the liquid phase falls within the reported values in the literature, as evident from Table 3. Additionally, d_{Si-Si}, Si−Si bond length, being an average distance between two bonded Si atoms predicted by ReaxFF$_{present}$ shows good agreement with the literature value of α-Si. From the graph in Figure 7, partial radial distribution functions (RDFs) of liquid and solid phases generated by both versions of the force field perfectly match each other.

2.3.2. Validation of Oxygen Diffusion in Silicon. To reinforce the results presented in Section 2.2.4 that provides a clear indication of O-interstitial migration mechanism in bulk Si reproduced by ReaxFF$_{present}$, the diffusion process was also addressed at different target temperatures within the range of (800−2400 K) by applying ReaxFF$_{SiOH(2010)}$ and ReaxFF$_{present}$. For this aim, a supercell containing 512 Si atoms was generated from a diamond Si unit cell with a dimension of 5.463 Å, and 15 O atoms were randomly located at the bond centers of the 2 nearest-neighbor host atoms in the crystal. Unfolding trajectories of O atoms in the bulk Si were achieved at different temperatures in for only two steps: in the first step, the system was gradually annealed at target temperature in an NVT ensemble using a Berendsen thermostat with a coupling constant of 100 fs for 1 ns. After reaching equilibrium at the target temperature, the system was maintained for 3.5 ns with a coupling constant of 1000 fs by approaching NVE ensemble to achieve a fairly linear mean square displacement (MSD) plot. The diffusion trajectories were collected every 50 fs for 3.5 ns. Time steps were set at 0.5 fs. The diffusion coefficient describing the diffusion behavior of the particle, $D(t)$, was computed using the Einstein diffusion relation[49]

$$D(t) = \frac{1}{2d} \lim_{\Delta t \to \inf} \frac{\mathrm{MSD}(t + \Delta t) - \mathrm{MSD}(t)}{\Delta t} \tag{5}$$

where d is the dimension, MSD(t) is the evolution of particle motion with respect to the reference particle and calculated by the following equation

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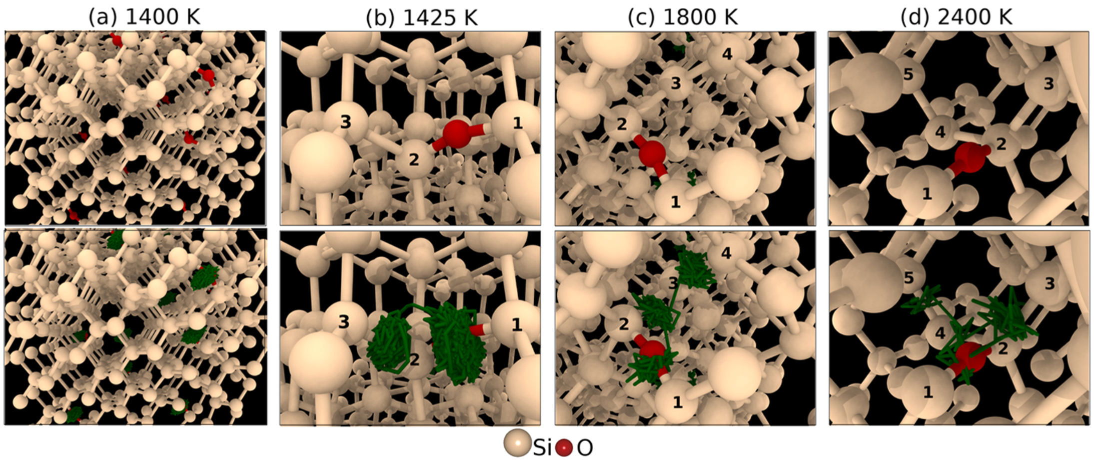

Figure 8. Snapshots taken from the final MD steps of the simulations by ReaxFF$_{present}$ at different target temperatures. The top four snapshots represent the initial configuration of the system from different views, and the bottom four images show the diffusion trajectory of O atom for 3.5 ns where the green line represents the atom's trajectory. (a) No diffusion was observed at 1400 K, and O atoms at BC sites vibrate around themselves. (b) At 1425 K, O atom residing in the BC site neighboring 1 and 2 Si atoms hop to the next BC site between 2 and 3 Si atoms, and, during the jumping, O atom goes through the transition state at the S site located above 2 Si atom. (c) At 1800 K, O atom initially being at BC site neighboring 1 and 2 atoms, first, jumps to the next BC site between 2 and 3 Si atoms, and then hops the other BC site between 3 and 4 Si atoms. (d) At 2400 K, O atom diffuses by hoping, first, the BC site between 1 and 2 Si atoms followed by the BC sites between 2 and 3 Si atoms, and 3 and 4 Si atoms. Finally, the O atom occupies the BC site between 4 and 5 Si atoms (OVITO[40]).

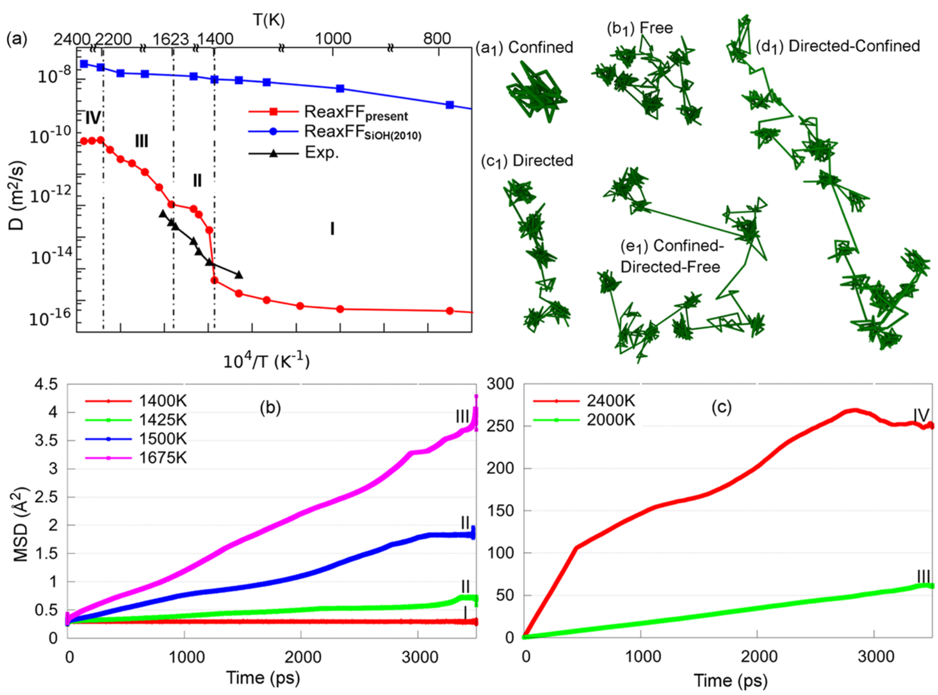

Figure 9. (a) Temperature dependence of the diffusion coefficient computed by both versions of the ReaxFF force field and experiment,[50] (b, c) MSD profiles at the given temperatures where the numeration with roman numbers refers to indicated regions in (a). The observed trajectory types of O atom during diffusion are as follows: (a_1) confined, (b_1) free, (c_1) directed, and anomalous diffusion types combined of (d_1) directed-confined and (e_1) directed-free-confined. The main difference between the trajectories in (d_1) and (e_1) is that O atoms intermittently move away from their initial locations in (d_1) contrary to O atoms moving back toward their initial positions in (e_1). Note that the experimental values of the diffusion coefficient displayed in the graph in (a) were obtained by using a normalized diffusion coefficient equation provided by ref 50.

$$\mathrm{MSD}(t) = \langle |r(t) - r(0)|^2 \rangle \tag{6}$$

and t is the time. In addition, the relationship between temperature and diffusion coefficient averaged over all O atoms in Si was established.

Figure 8 shows the diffusion trajectories of O atom generated by ReaxFF$_{present}$ at different temperatures such 1400, 1425, 1800, and 2400 K, which is a clear indication that ReaxFF$_{present}$ successfully directed every O atom to occupy the BC sites regardless of target temperature, as reported in DFT studies.[9−11] Figure 8a also shows that O atoms are confined to vibrate around themselves and cannot hop the other BC sites during the simulation until above 1400 K because they do not have energy required for overcoming the barrier. They reach the transition state and jump to the next BC site at temperatures over 1400 K (Figure 8b). At further heat

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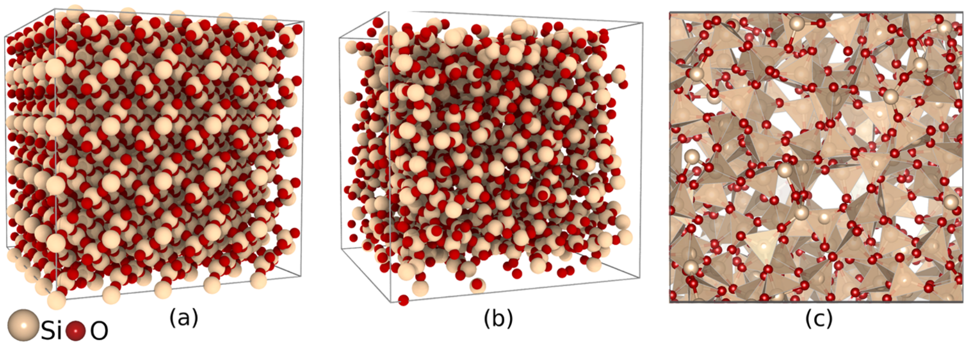

Figure 10. (a) Initial bulk configuration of crystalline β-SiO_2, (b) a-SiO_2 quenched from the liquid state to 300 K, and (c) the polyhedral model of a-SiO_2 (VESTA[39]).

treatment, $ReaxFF_{present}$ successfully allows O atoms to still hop between only the two nearest BC sites even at higher temperatures below the computed melting point of bulk Si (Figure 8c,d).

Figure 9a shows the comparison of $ReaxFF_{present}$, $ReaxFF_{SiOH(2010)}$, and the experimental result[50] in terms of the temperature dependence of the diffusion coefficient. The new model of force field predicts the diffusion coefficient in overall good agreement with the experimental value.[50]

We also examined the diffusion behavior of O atom based on $ReaxFF_{present}$ within the range of (800−2400 K). As illustrated in Figure 9a, the plot belonging $ReaxFF_{present}$ reveals four different regions depending on the temperature. The distinct character of each region is illustrated by the MSD plots labeled with the region numbers in Figure 9b,c and the representative diffusion trajectories in Figure 9a_1−e_1. In region I between 800 and 1400 K, the diffusion does not occur where O atoms are confined to vibrate around themselves (Figure 9a_1 and MSD plot labeled with I in Figure 9b). In the other regions, O atom exhibits anomalous diffusion behavior in different combinations of confined (Figure 9a_1), free (Figure 9b_1), and directed (Figure 9c_1) diffusion types.[51] In region II, the slopes of MSD curves labeled with II in Figure 9b decrease and finally, the MSD profiles reach a plateau. This means that O atoms, first, begin to freely diffuse through jumping their neighboring BC sites at temperatures over 1400 K, and then, are likely blocked by their neighbors and so begin to exhibit a confinement trajectory (Figure 9a_1) around themselves. In region III, O atoms are energetically free to further diffuse between the BC sites. The MSD plot representing region III in Figure 9b exhibits a combination of likely directed and confined as illustrated in Figure 9d_1 where O atoms intermittently move away from their initial points in an accelerated manner. The MSD plot, again, labeled with III in Figure 9c shows a predominantly free (normal) diffusion. In region IV, MSD plots generally exhibit the downward trend in some places toward to the end as shown in Figure 9c, likely caused by the accelerated O atoms hitting the other atoms with rising temperature and being directed toward their initial frames by changing the direction (Figure 9e_1).

2.3.3. Silica Glass Using Melt-Quenching Method. With a particular focus on a-SiO_2/Si system at high temperatures, first, the a-SiO_2 was prepared by applying the melt-quenching method with the application of $ReaxFF_{SiOH(2010)}$ and $ReaxFF_{present}$, separately. The initial configuration of bulk crystal SiO_2 (c-SiO_2), including 1000 Si and 2000 O atoms, was built from the β-cristobalite unit cell with a lattice constant of 7.13 Å[52] and established in a cubic simulation box with a dimension of 35.65 Å with periodic boundary conditions. The space group and space group number of the β-cristobalite SiO_2 are $Fd\bar{3}m$ and 227, respectively. The MD energy minimization was first performed to avoid the overlapped atoms which lead to excessive forces and energy differences between consecutive steps in MD. The relaxed system was annealed twice from 7000 to 300 K in NVT and NPT ensembles, separately, at zero pressure to achieve a glass having correct structural properties. In the first annealing process, the initial temperature of the bulk c-SiO_2 was set at 7000 K, and the bulk was equilibrated at such high temperature for 50 ps in an NVT ensemble until achieving a completely liquid phase. Subsequently, the system was cooled down 300 K in 100 ps and relaxed at 300 K for 500 ps to reduce the density of intrinsic point defects located in the structure. In the second annealing process, the system was again heated at 7000 K for 100 ps and cooled down 300 K for 200 ps, in this case, in an NPT ensemble. Finally, the amorphous structure was relaxed in the NPT ensemble for 100 ps. The relaxation resulted in a 39.7508 × 39.0886 × 36.7919 $Å^3$ glass. In the simulation, a velocity Verlet algorithm was employed to integrate Newton equation of motion, and the time step is 0.5 fs. The temperature-damping and the pressure-damping constants were set at 100 and 5000 fs, respectively. The validation of the glasses prepared by the $ReaxFF_{SiOH(2010)}$ and $ReaxFF_{present}$ was performed against DFT/experimental data using their computed structural properties including mass density, partial RDFs, coordination number distribution, and bond angle distributions (BADs).

Figure 10 shows the initial configuration of bulk c-SiO_2 and its amorphous form obtained by applying the new model of ReaxFF. $ReaxFF_{present}$ predicts the mass density of the glass ($glass_{present}$) to be 2.21 gr/cm^3, showing an excellent agreement with the experimental value of 2.20 gr/cm^3 [53,54] (Table 4).

Figure 11a and Table 4 show that the average Si−O bond distances (d_{SiO}) computed by $ReaxFF_{SiOH(2010)}$ and $ReaxFF_{present}$ are in good agreement with each other and the experimental values. The Si−Si bond distance, 3.1 Å, predicted by the $ReaxFF_{SiOH(2010)}$, is nearly the same as experimental bond distance,[53] and the new force field model yields a slight narrowing in the Si−Si bond distance. For O−O bond distance, $ReaxFF_{present}$ is in a good agreement with the experiments.[53,55] From the data presented in Table 4 and the graph in Figure 11b, the new force field framework correctly generates Si−O−Si and O−Si−O angle distributions having peaks at 152 and 108°, respectively, being consistent with the experimental values. However, $ReaxFF_{SiOH(2010)}$ predicts that most of the Si−O−Si angles are centered around 154° where there is a slight difference of 3° from the

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Table 4. Structural Properties of the Silica Prepared by ReaxFF$_{SiOH(2010)}$ and ReaxFF$_{present}$

property	ReaxFF$_{SiOH(2010)}$	ReaxFF$_{present}$	experiment
density (gr/cm^3)	2.25	2.21	2.20[a,b]
d_{SiO} (Å)	1.59 ± 0.08	1.58 ± 0.09	1.608 ± 0.04[c]
Si–Si bond dist. (Å)	3.07 ± 0.14	3.02 ± 0.13	
O–O dist. (Å)	2.56 ± 0.32	2.58 ± 0.27	2.65[a]
Si–O RDF first max. (Å)	1.58	1.56	1.608[c], 1.620[a]
Si–Si RDF first max. (Å)	3.11	3.03	3.077, 3.12[a]
O–O RDF first max. (Å)	2.39	2.52	2.626[c], 2.65[a]
Si–O–Si (deg)	154.0	152.0	151.0[d,e,f]
O–Si–O (deg)	107.0	108.0	109.4[g], 109.5[a]
Si coordination	4.010	4.010	
O coordination	2.005	2.005	

[a]Ref 53. [b]Ref 54. [c]Ref 55. [d]Ref 56. [e]Ref 57. [f]Ref 58. [g]Ref 59.

experimental values.[56−58] Also, most of the O–Si–O angles in the glass$_{SiOH(2010)}$ gather around the 105°, and the analysis gives a peak at 107° with a shift of 2° from the experimental value. Both glasses have Si–O–Si and O–Si–O angular distributions in wide ranges contrary to that of the crystalline phase, which is a fundamental way of distinguishing the crystalline and the amorphous phases.[53]

Further analysis of the glasses performed in terms of the coordination number distribution is illustrated in Figure 11c. It shows that both glasses have similar coordination characteristics. Each of the glasses contains mostly 4-coordinate Si and 2-coordinate O atoms. Only 2 O and 9 O atoms have one and three Si neighbors, respectively, in the glass$_{present}$, and 4 O and 5 O atoms have one and three Si neighbors, respectively, in the glass$_{SiOH(2010)}$, despite their total bond order of 2. Moreover, as evident from the graph displayed in Figure 11c, there is no O–O radical remaining in the glasses. Around 96% of Si atoms in both glasses are 4-coordinate with the average Si–O bond order of 3.8. Figure 10c shows the 4-coordinate Si atoms in the polyhedral model of the glass$_{present}$ where the centers of the four nearest-neighbor O atoms around Si are joined by the lines. 3-Coordinate and 5-coordinate atoms constitute of 1.5 and 2.5% of Si atoms in the glass$_{present}$ with the bond orders of 2.81 and 3.84, respectively, and 1.2 and 2.1% of Si atoms in the glass$_{SiOH(2010)}$ with the bond orders of 2.88 and 3.88, respectively. In addition, only one and two Si–Si radicals were observed in the glass$_{present}$ and the glass$_{SiOH(2010)}$, respectively. All these structural characterizations belonging to the glass$_{present}$ and the glass$_{SiOH(2010)}$ prove that ReaxFF$_{present}$ successfully generates a glass from its crystalline form in good overall agreement as DFT and experiments.

2.3.4. a-SiO$_2$/Si System at High Temperatures. To address O diffusion directly in a-SiO$_2$/Si system, the lattice model of the system with dimensions of 65 × 65 × 80 Å^3 was constructed using the glass mentioned above and bulk crystal Si. To create a relatively larger glass, the coordinates of a-SiO$_2$ were replicated and translated in the *y*-direction and adjacent to the original ones. a-SiO$_2$ glass containing 5802 atoms in dimensions of 65 × 65 × 22 Å^3 was cut from the generated larger glass. After that the energy minimization was individually performed to the cut a-SiO$_2$ glass to prevent the overlapping atoms and bad contacts. The relaxed a-SiO$_2$ was then deposited on the diamond Si matrix normal to *z*-axis by leaving a gap of 2 Å between two layers, and finally, a-SiO$_2$/Si system comprising 14 754 atoms in the dimensions of 65 × 65 × 100 Å^3 was constructed by inserting a vacuum region with a height of 20 Å at the top of the silica. With the application of ReaxFF$_{SiOH(2010)}$ and ReaxFF$_{present}$, separately, the total system was annealed at temperatures ranging between (300 and 2400 K) for 500 ps.

Figure 12 illustrates the MD snapshots of the a-SiO$_2$/Si system annealed based on ReaxFF$_{SiOH(2010)}$ and ReaxFF$_{present}$. The plot displayed in Figure 13 also shows the solubility and

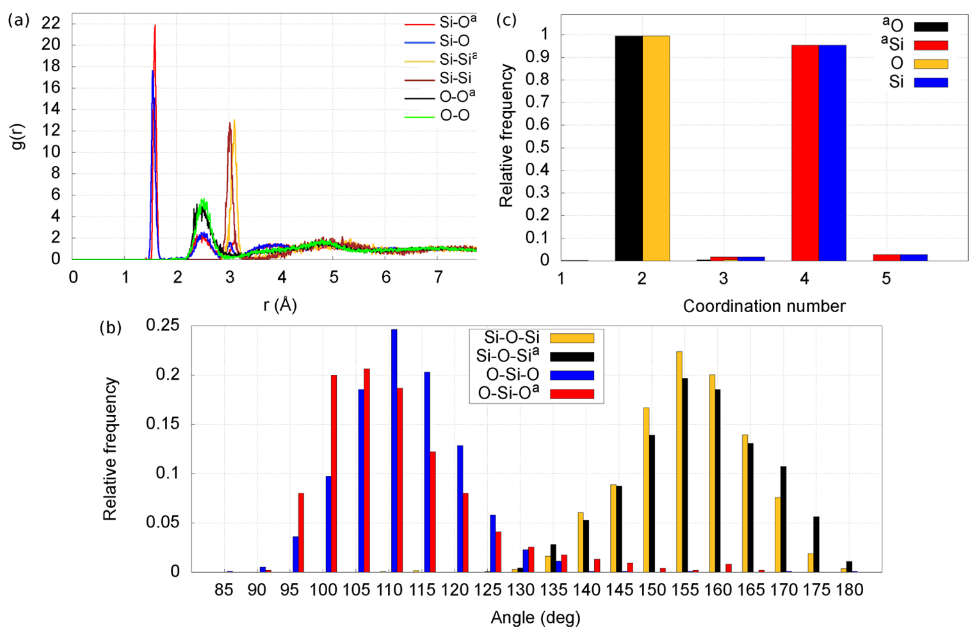

Figure 11. Structural characterizations of glass$_{present}$ and glass$_{SiOH(2010)}$, where the results glass$_{SiOH(2010)}$ are distinguished with the superscript a from those of glass$_{present}$. (a) Si–O, Si–Si, and O–O partial RDFs, (b) O–Si–O and Si–O–Si BADs, and (c) analysis for coordination number distribution of both glasses quenched from the liquid state to 300 K.

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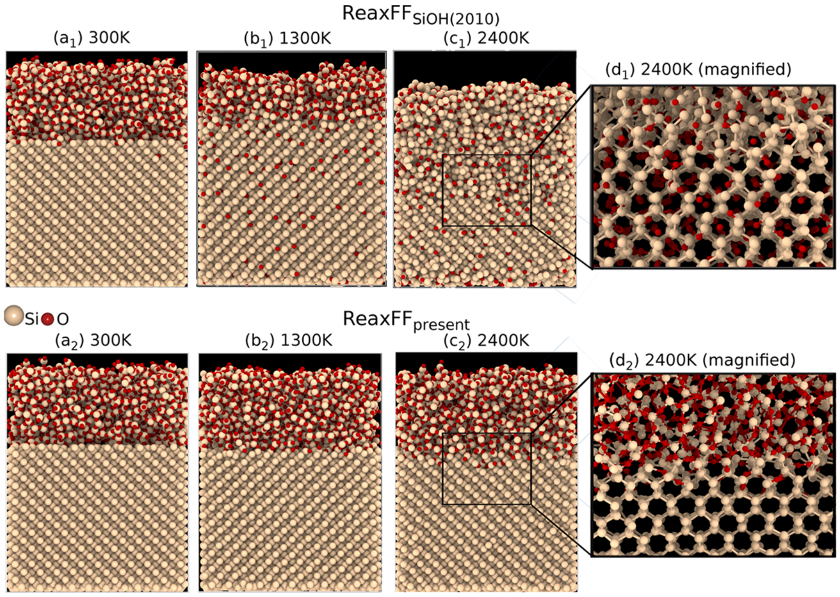

Figure 12. a-SiO_2/Si system annealed at temperatures ranging between (300 and 2400 K) for 500 ps by applying ReaxFF$_{present}$ and ReaxFF$_{SiOH(2010)}$. Snapshots shown in (a$_1$)−(d$_1$) belong to the results generated by ReaxFF$_{SiOH(2010)}$. (a$_2$)−(d$_2$) represent the results reproduced by ReaxFF$_{present}$. Also, (d$_1$) and (d$_2$) illustrate the magnified views of the framed regions with black enclosures at 2400 K (OVITO[40]).

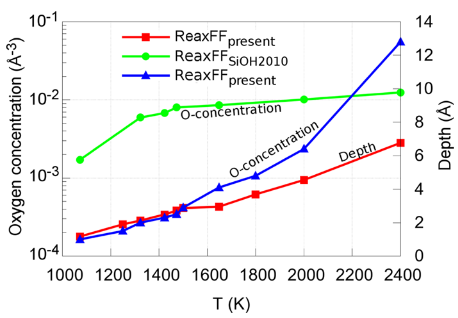

Figure 13. Temperature dependence of O solubility and depth profiles in the Si substrate based on ReaxFF$_{present}$ and ReaxFF$_{SiOH(2010)}$ under heat treatment where the label of ReaxFF$_{present(depth)}$ represents the depth profiling of O generated by the new force field, and the labels of ReaxFF$_{present}$ and ReaxFF$_{SiOH(2010)}$ belong to the O−concentration plots.

the depth distribution of O-interstitial in the Si substrate depending on the annealing temperature. Note that the depth profiling based on ReaxFF$_{SiOH(2010)}$ was not shown in the graph in Figure 13 since O atoms coming from the silica layer were determined throughout the Si substrate at the temperatures within the range of (1300−2400 K), as evident from Figure 12a$_2$. ReaxFF$_{SiOH(2010)}$ resulted in O diffusion even at low temperature such as 300 K (Figure 12a$_1$), whereas ReaxFF$_{present}$ succesfully resulted that the O atoms react with only the bonds between the Si atoms in the first and second layer of the Si substrate in the vicinity of the interface where O-depth in the Si matrix is about 2 Å, until 1400 K (Figure 13). At temperatures over 1400 K, O atoms begin to jump the sites neighboring the Si atoms located in the third layer of the Si substrate. The most striking difference between both versions of the ReaxFF force fields can be explicitly seen from Figure 12d$_1$−d$_2$. In Figure 12d$_2$, ReaxFF$_{present}$ successfully directs the O atoms to occupying the BC sites in the Si matrix even at 2400 K, being consistent with DFT/experimental results,[9−12,36] whereas ReaxFF$_{SiOH(2010)}$ results in positioning of O atoms at the H and T sites inside of honeycomb units through the Si substrate instead of the BC sites, as can be seen in Figure 12d$_1$.

3. CONCLUSIONS

In the present study, the major driving force behind the improvement of ReaxFF$_{SiOH(2010)}$[18] is to ensure the MD simulations of a-SiO_2/Si interface event at high temperatures below computed melting point of bulk Si. For this purpose, we refitted the ReaxFF$_{SiOH(2010)}$ force field parameters by putting a special emphasis on O diffusion in bulk Si and O-related point defect formations in the vicinity of a-SiO_2/Si interface. A thorough optimization resulted that the formation energy of O-vacancy in the silica agrees with DFT.[32] Additionally, the new force field framework predicts the order of relative stability of O-interstitial sites from most stable to least stable to be BC → S → T → H, showing good agreement with ref 9. Moreover, ReaxFF$_{present}$ correctly describes the asymmetric diffusion mechanism of O in Si network which occurs through the jumps between the neighboring BC sites by going through the transition state at the split site,[9−11,36] which is the most remarkable conclusion of this work. The diffusion barrier is predicted to be 64.8 kcal/mol, being reasonably close to the experimental reported value of 58.3 kcal/mol.[13,14] We further validated ReaxFF$_{present}$ force field parameters by performing the MD simulations in many aspects: in the first step, the melting point of bulk Si was determined by applying two-phase approach, and the structural properties of the liquid and solid phases of Si were analyzed based on the mass density and the pair correlation function. Second, the O diffusion behavior in Si matrix was investigated at different target temperatures until below the computed melting point of Si, and the temperature dependence of the diffusion coefficient was established. Our results demonstrate that ReaxFF$_{present}$ allows O atoms to diffuse through the Si matrix at temperatures over 1400 K. The relationship between the temperature and diffusion coefficient produced by ReaxFF$_{present}$ fits in well with the experiment.[50] We also reported that O atoms exhibit the anomalous diffusion

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behavior in different combinations of free, directed, and confined diffusion. As the last step of the force field validation, a-SiO_2/Si system was modeled by using the glass with a mass density of 2.21 gr/cm^3, showing good agreement with the reported experimental value of 2.20 gr/cm^3,[53] and simulated at temperatures within the range of (300−2400 K). We demonstrated that $ReaxFF_{present}$ succeeded to overcome the diffusion problem observed by $ReaxFF_{SiOH(2010)}$ even at low temperatures, and reproduced experimentally/theoretically defined O-interstitial-mediated diffusion mechanism in the system as a consequence of improving the force field with a particular emphasis on the formation energies of the O-interstitial sites and the diffusion pathway of O atom in the bulk Si. As such, we believe that the new force field description for Si/O/H interactions provides an important computational tool for studying SiO_2/Si interfaces.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpca.9b01481.

$ReaxFF_{present}$ and $ReaxFF_{SiOH(2010)}$ force field parameters for Si/O/H interactions. DFT and ReaxFF calculations for equations of state of silica and silicon phases (PDF)

"geo" - the non-periodic and periodic structures incorporated in the force field training set (geo), "trainset.in" - the QM-data set, and "fort.99" - a detailed cost-function report based on the ReaxFF & QM values comparison (ZIP)

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Notes

The authors declare no competing financial interest.

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